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Tableau Assignment

Assignment 1: Data Preparation and Integration

- Question:** Combine the datasets into a single unified dataset for analysis.
 - Join train.csv with features.csv on Store and Date.
 - Add store details from the stores.csv file by joining on Store.

Walmart Data Source

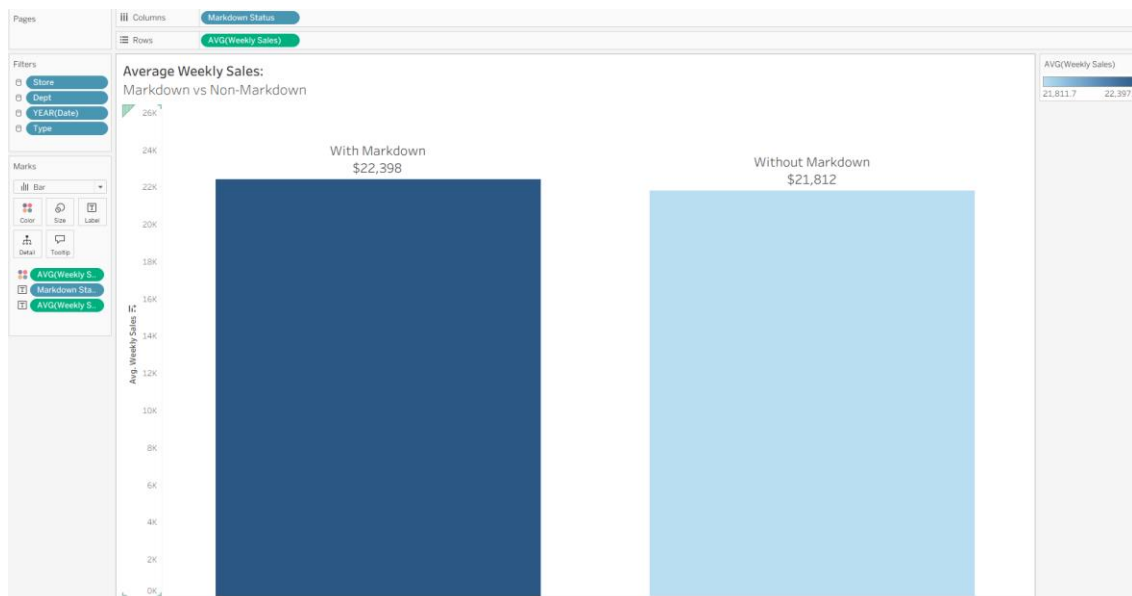
train.csv

features.csv

stores.csv

Name	train.csv
Store	train.csv
Dept	train.csv
Date	train.csv
Weekly Sales	train.csv
Is Holiday	train.csv
KPI Holiday vs Non Holiday	train.csv
Metric	train.csv
Sales Correlation	train.csv

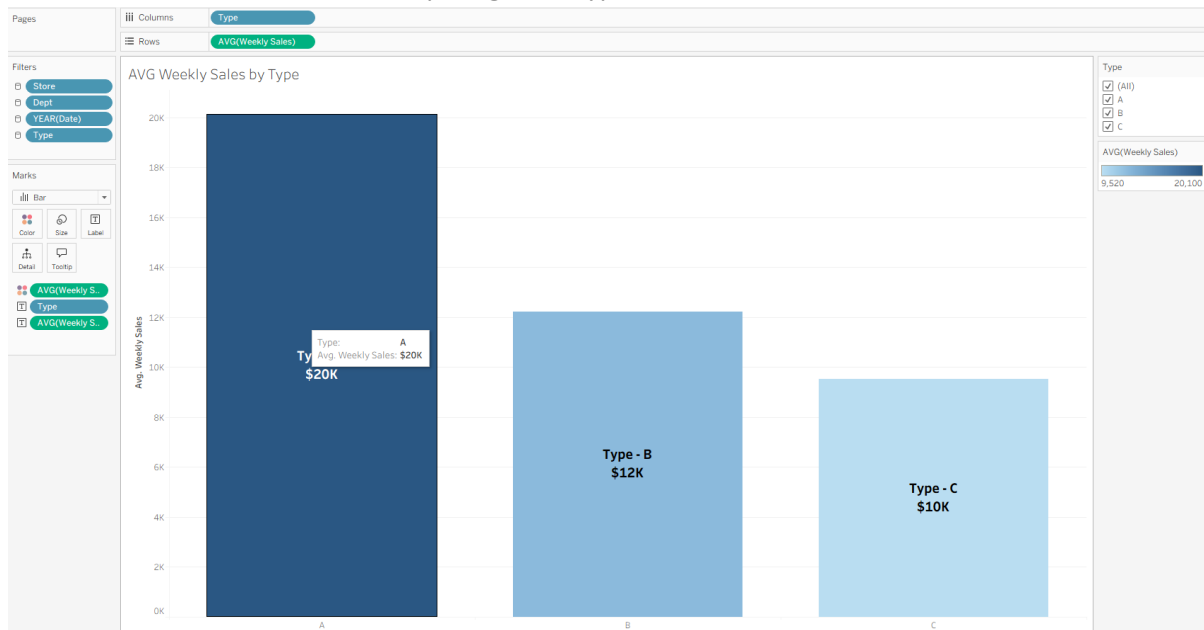
- Question:** Explore the relationship between weekly sales and markdowns.
 - Calculate the average weekly sales for rows with and without markdown data (Markdown1 to Markdown5).



Assignment 2: Store Performance Analysis

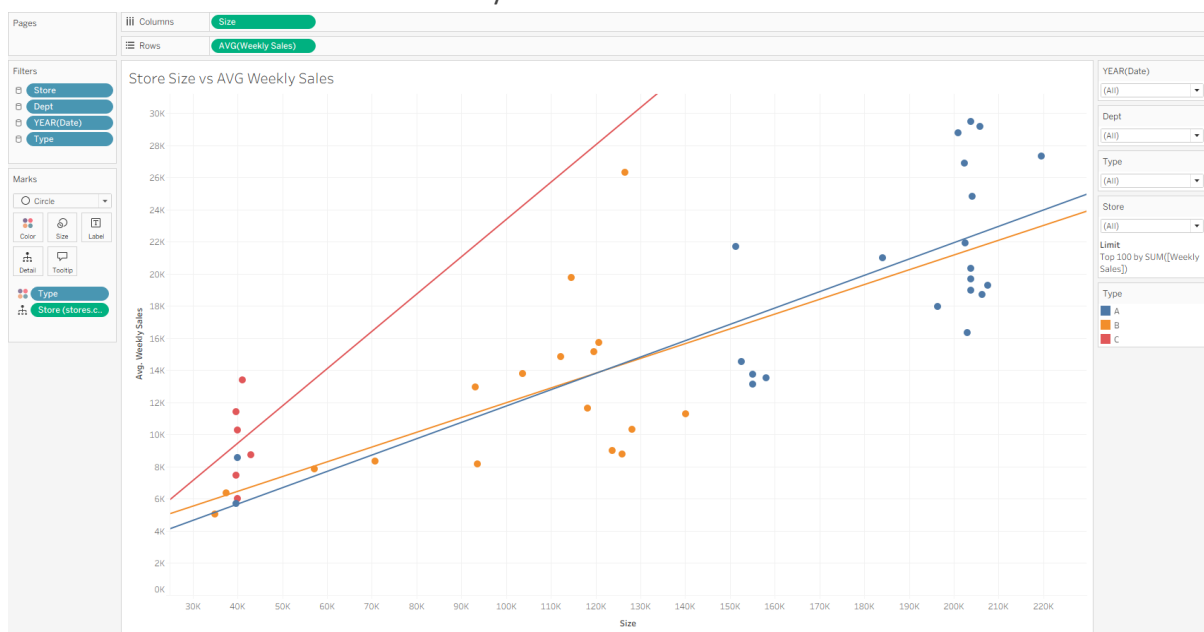
1. Question: Which store type (A, B, or C) has the highest average weekly sales?

- Create a chart comparing store types.



2. Question: How does store size influence weekly sales?

- Create a scatter plot with Size on the X-axis and average Weekly_Sales on the Y-axis.
- Add trendlines to identify correlations.

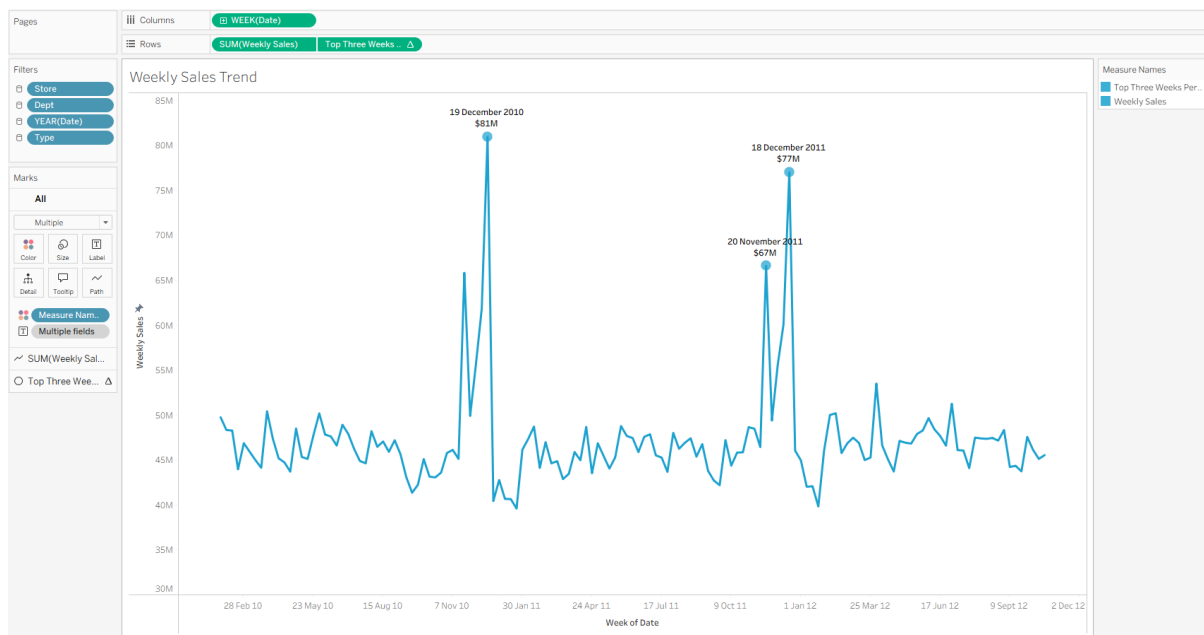


3. **Question:** Identify the top 5 stores by total sales.
 - o Create a chart to highlight the top-performing stores.



Assignment 3: Weekly Sales Trends

1. **Question:** Analyze weekly sales trends over time.
 - o Create a chart with Date and Weekly_Sales.
2. **Question:** Highlight spikes in sales.
 - a. Identify and annotate the top 3 weeks with the highest sales.

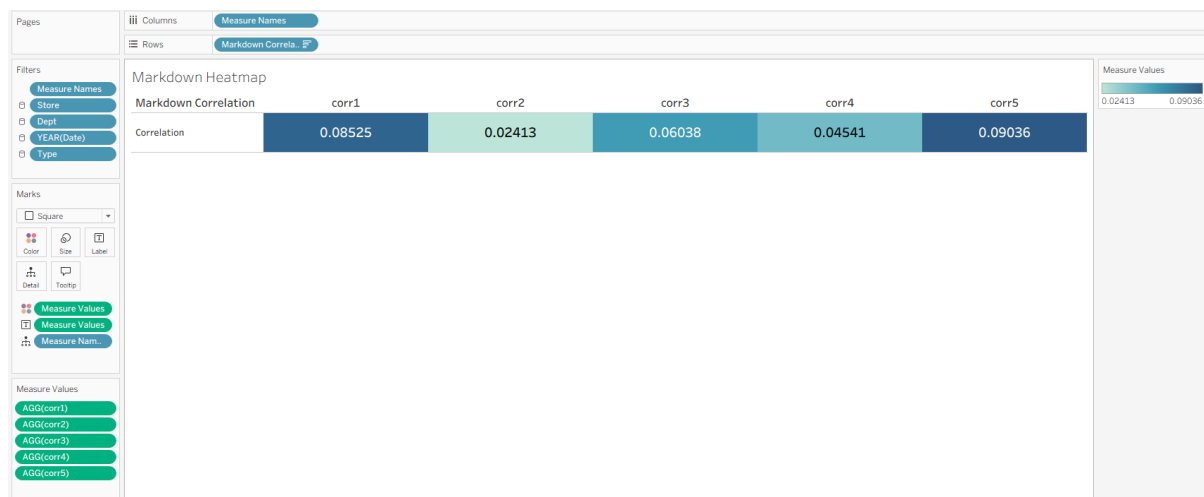


3. Question: Compare holiday vs. non-holiday sales



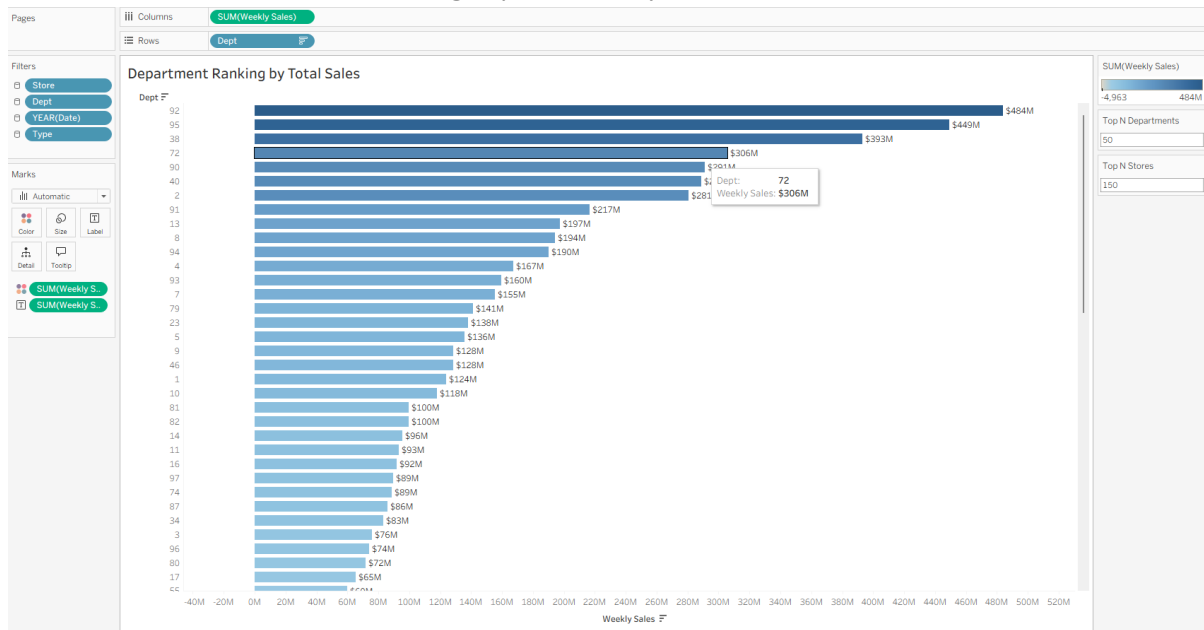
Assignment 4: Markdown Effectiveness

- Question:** Evaluate the impact of markdown events on weekly sales.
 - Create a heatmap showing the correlation between markdowns (MarkDown1 to MarkDown5) and sales.
- Question:** Which markdown event has the most significant impact?
 - Use calculated fields to compute and visualize the correlation coefficients.

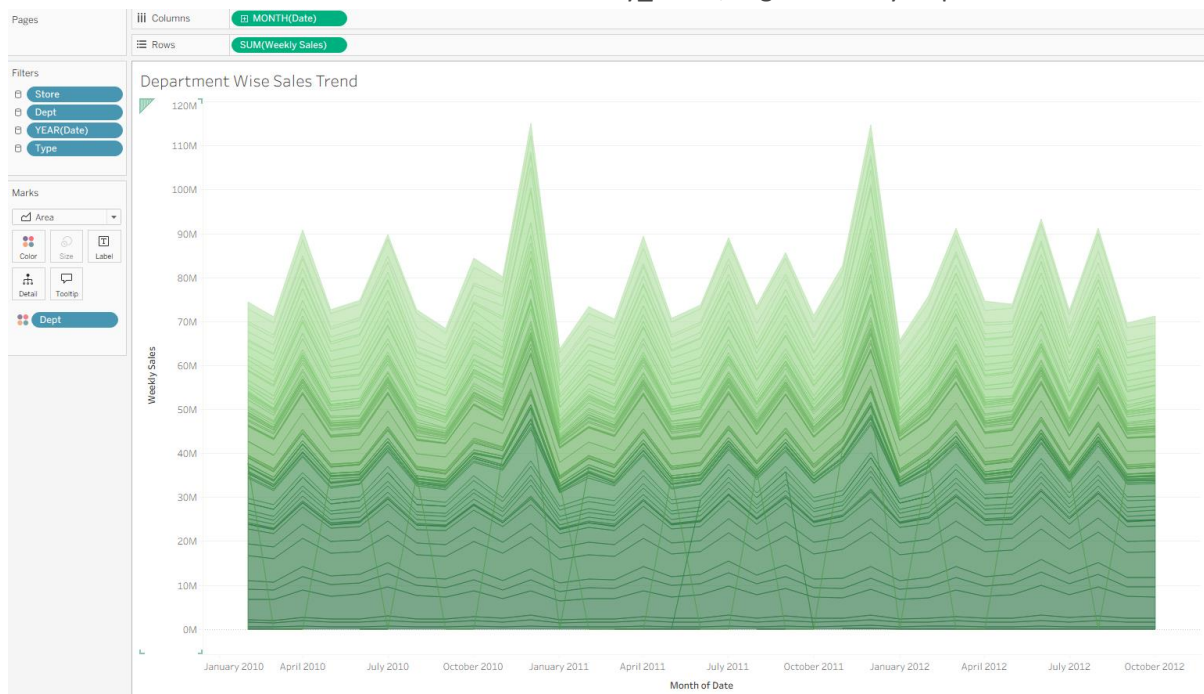


Assignment 5: Department-Level Analysis

1. **Question:** Which department generates the highest sales?
 - Create a chart ranking departments by total sales.



2. **Question:** Analyze department-wise sales trends.
 - Create a chart with Date and Weekly_Sales , segmented by Dept.

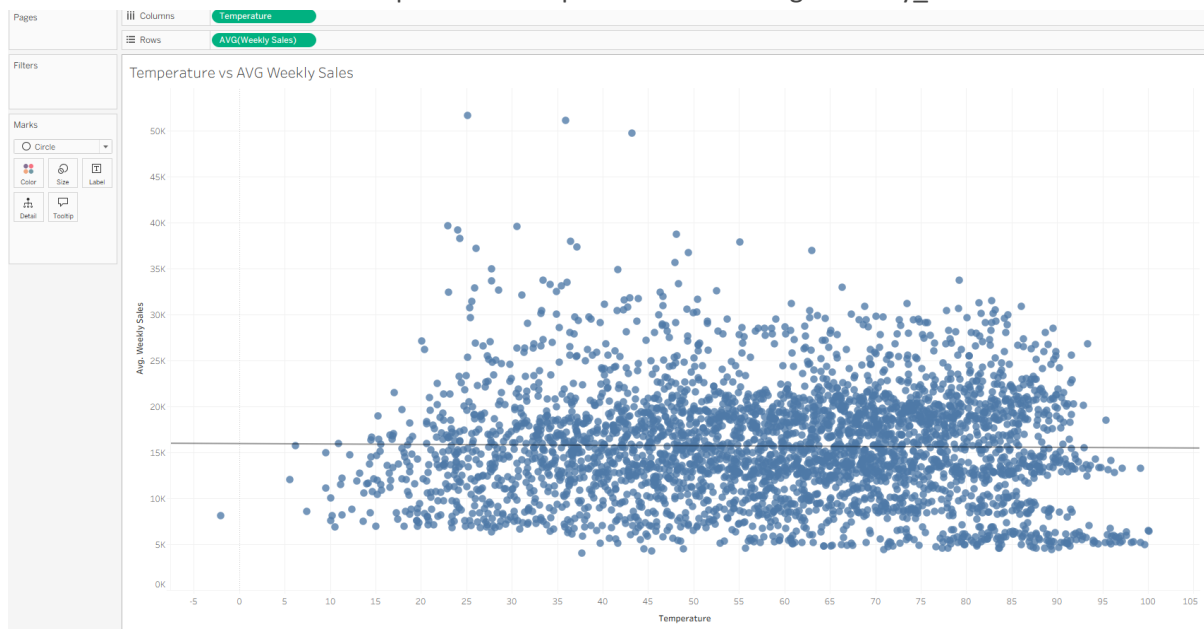


3. **Question:** Identify departments with significant holiday sales increases.
- Use filters or calculated fields to compare holiday vs. non-holiday sales.



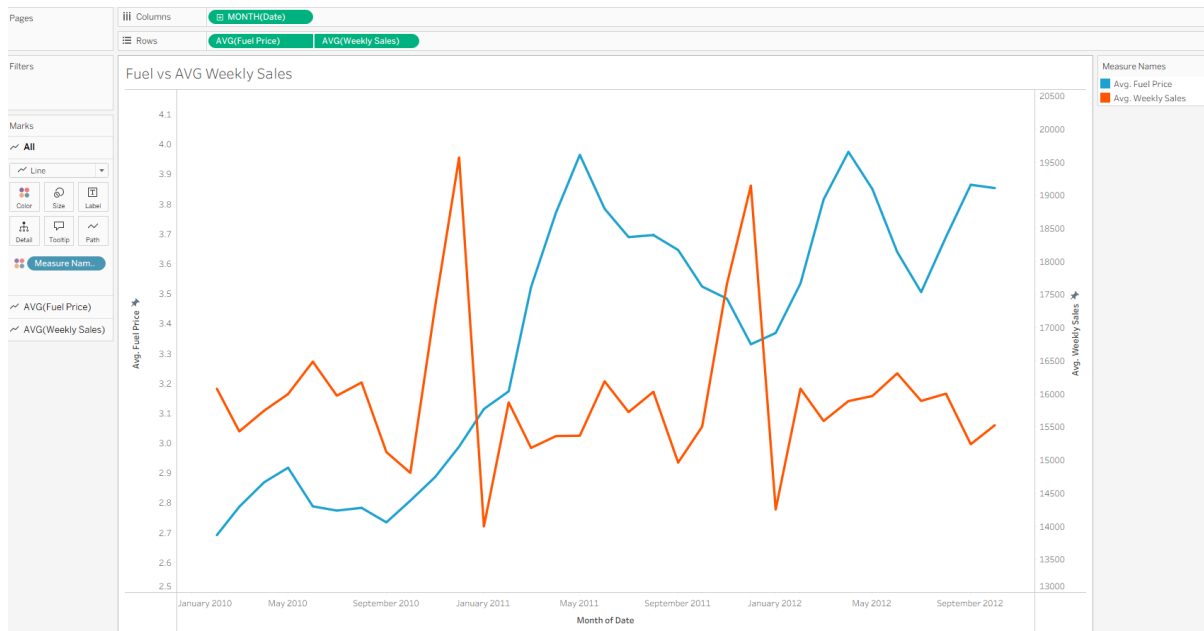
Assignment 6: Regional Analysis

1. **Question:** Analyze sales performance by temperature.
- Create a scatter plot with Temperature and average Weekly_Sales.



2. **Question:** Explore the effect of fuel prices on sales.

- Create a dual-axis chart showing Fuel_Price and Weekly_Sales trends over time.



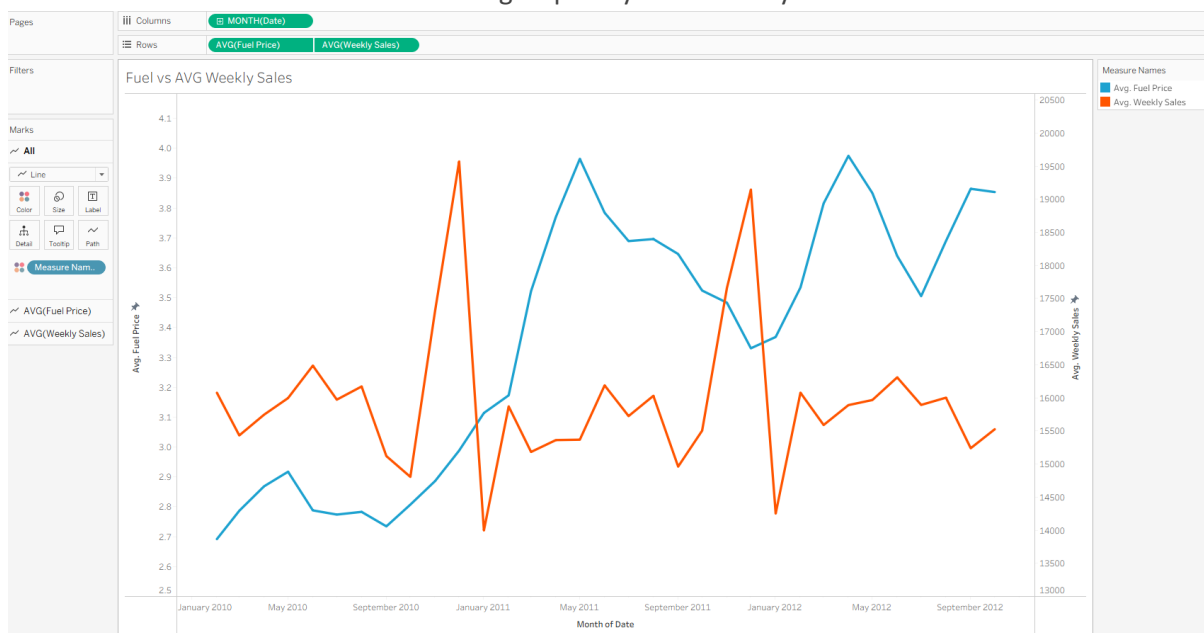
3. **Question:** Identify regions most affected by unemployment rates.

- Use a geographic map to display unemployment rates by store location.

Assignment 7: Seasonality and Holiday Analysis

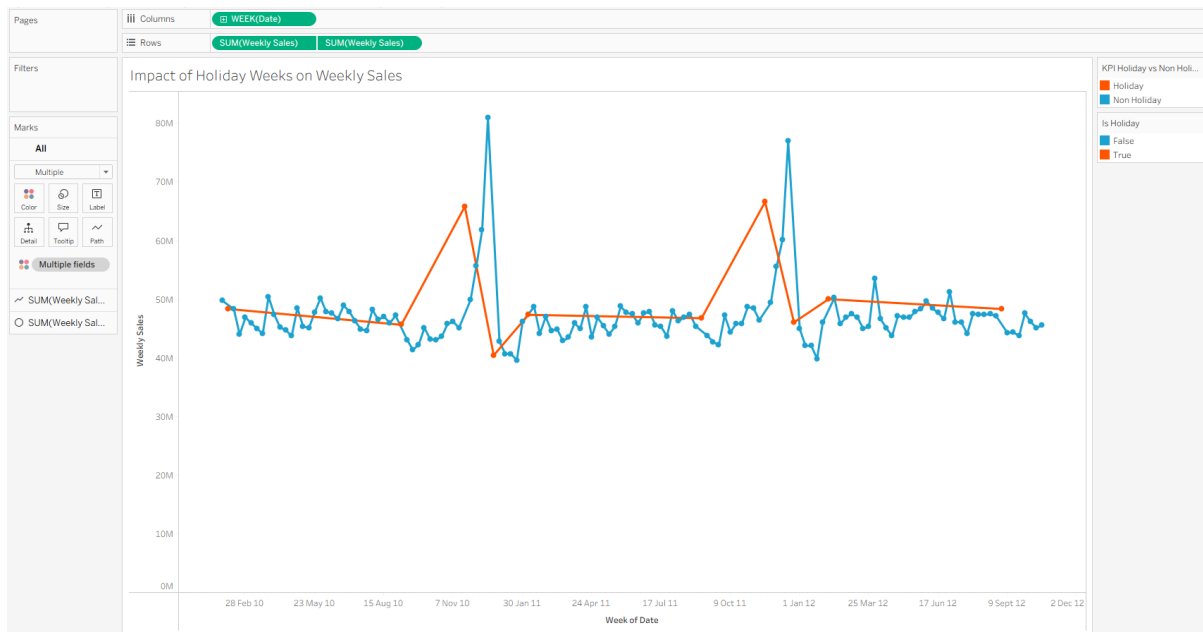
1. **Question:** Analyze sales seasonality over months and years.

- Create a chart with sales grouped by month and year.



2. Question: Do holidays drive significant sales spikes?

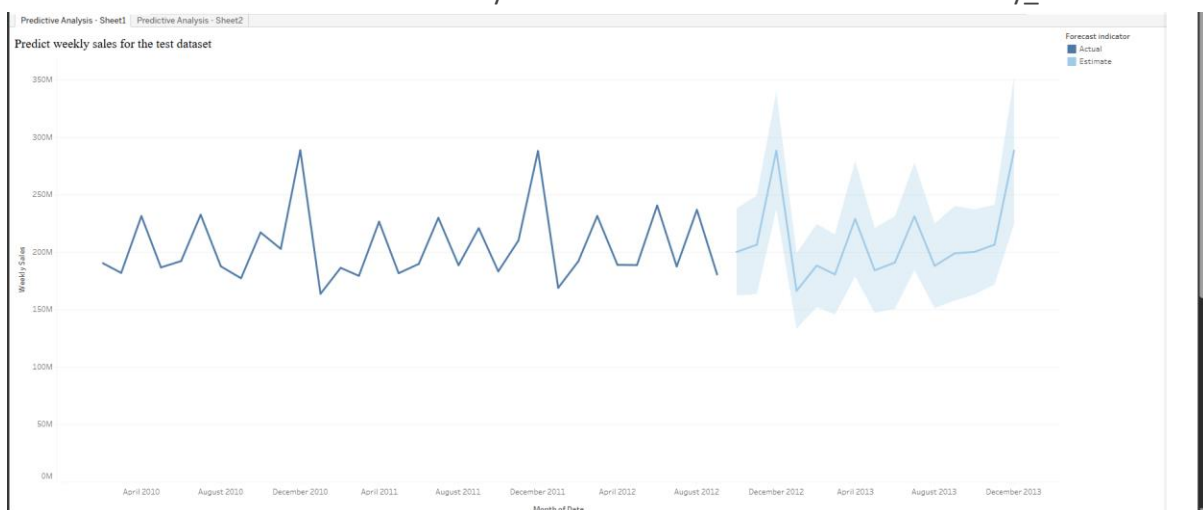
- Overlay holiday weeks on the line chart for a visual comparison.



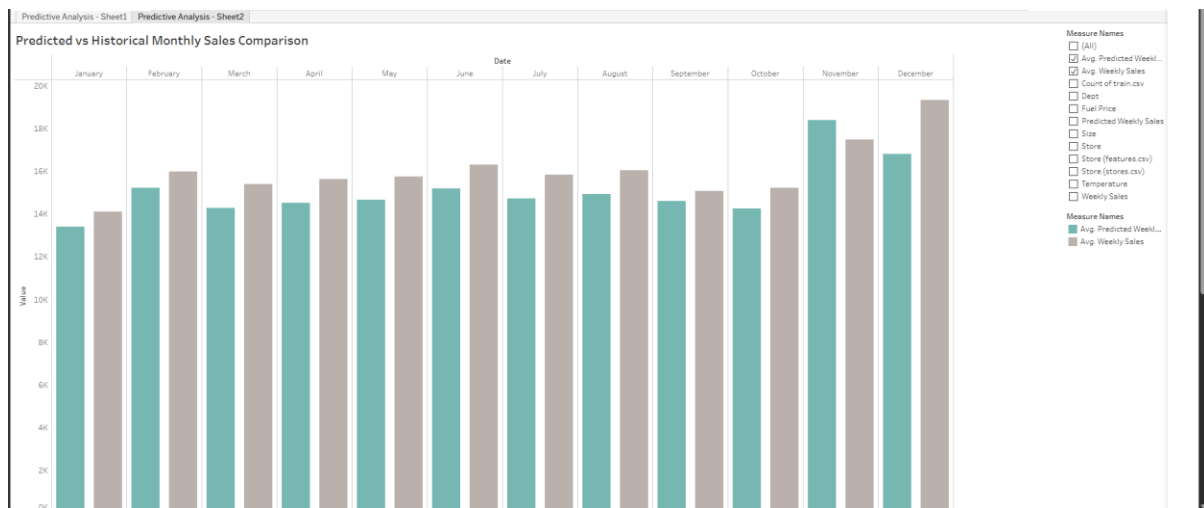
Assignment 8: Predictive Analysis (Test Data Integration)

1. Question: Predict weekly sales for the test dataset.

- Integrate test data (test.csv) with feature and store datasets.
- Use Tableau's trend analysis and calculated fields to estimate Weekly_Sales.

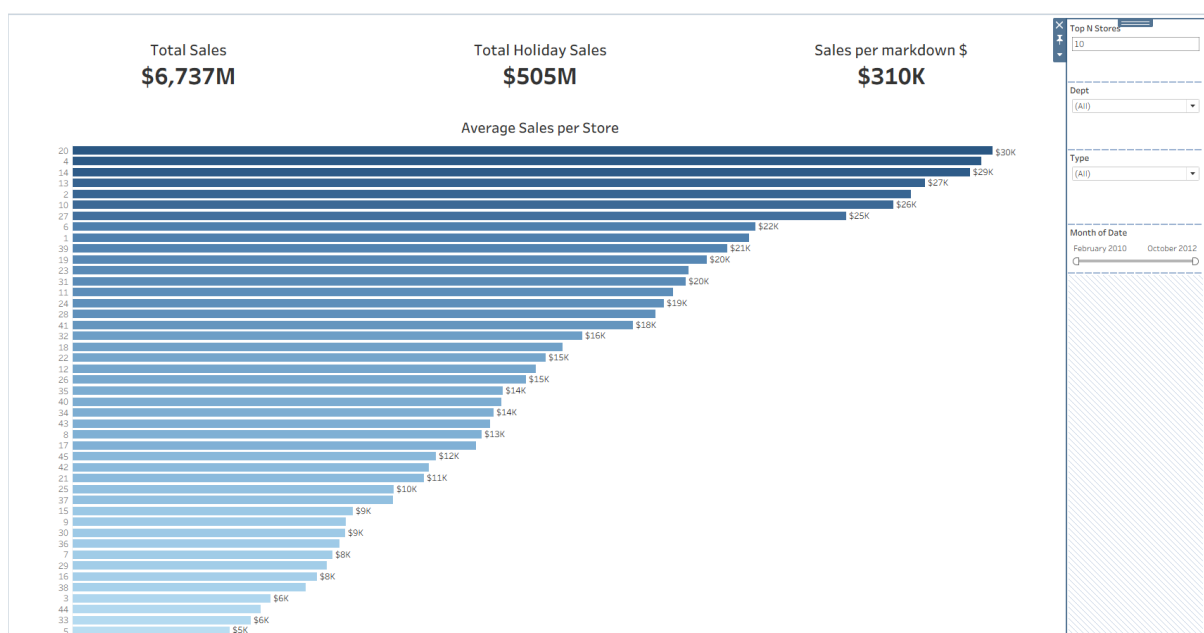


2. **Question:** Compare predicted sales to historical sales for similar time periods.
 - Create side-by-side bar charts to visualize differences.



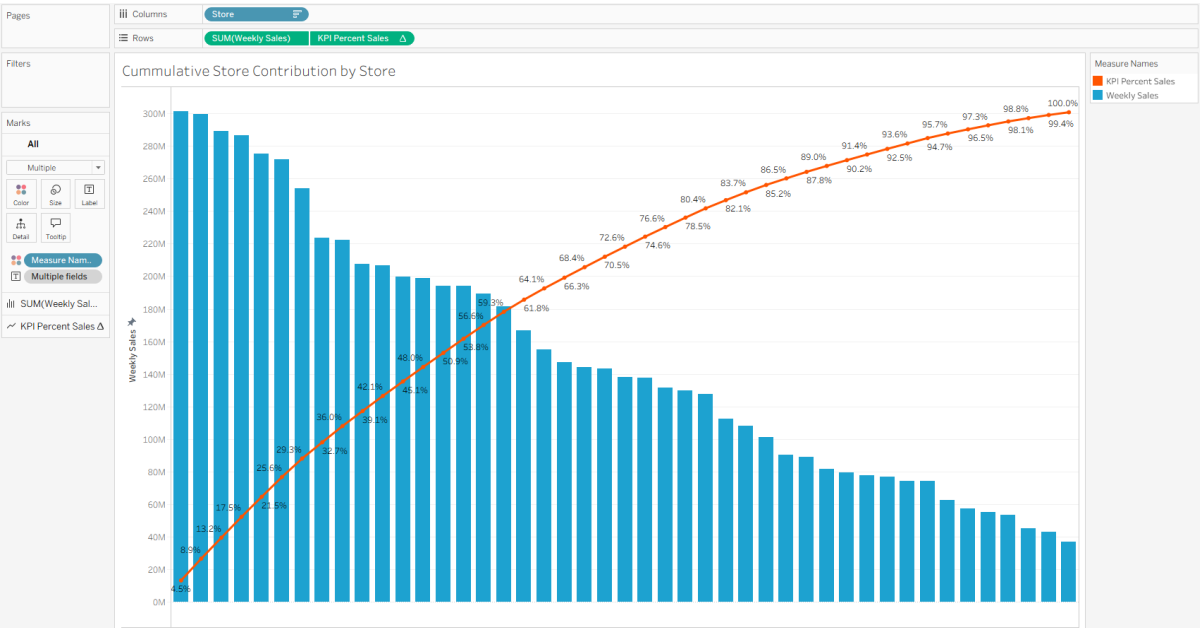
Assignment 9: KPI Dashboard Creation

1. **Question:** Create a dashboard to showcase key performance indicators (KPIs):
 - Total sales.
 - Average sales per store.
 - Total holiday sales.
 - Markdown effectiveness.
2. **Question:** Add interactivity to the dashboard.
 - Enable filters for store type, department, and date range.

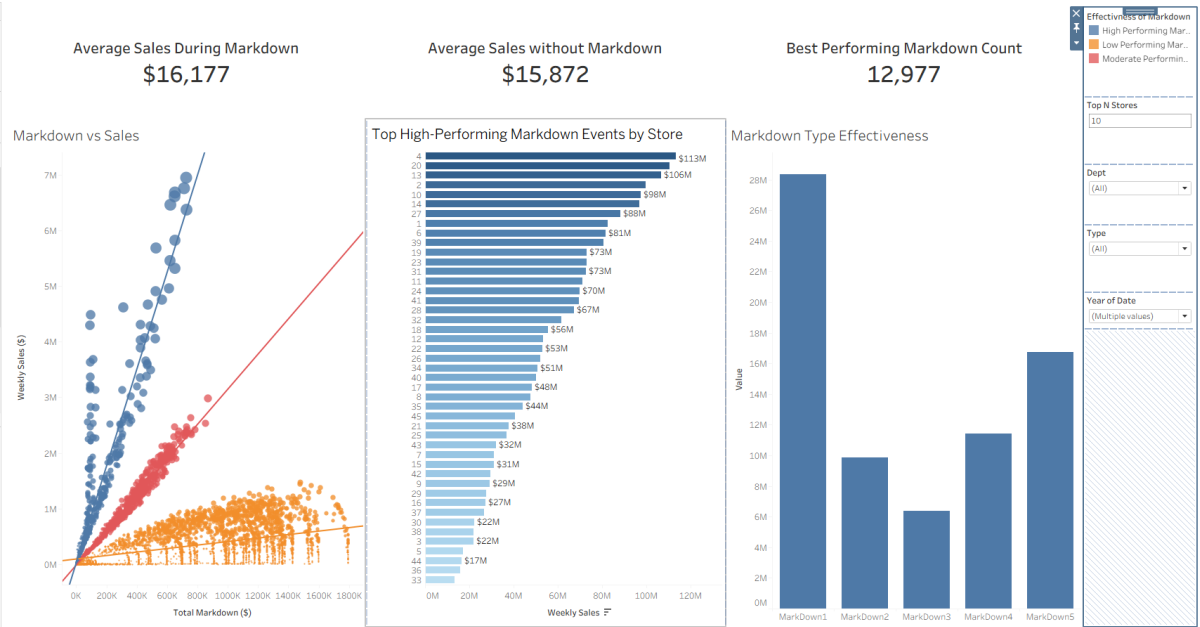


Assignment 10: Advanced Insights

1. Question: Create a chart to show cumulative sales contribution by store or department.
- o Identify the top contributors to sales.



2. Question: Perform a profitability analysis based on markdown effectiveness.
- o Create a dashboard to highlight high-performing markdown events.



3. **Question:** Design a "What-If" analysis to simulate sales scenarios based on changes in markdowns or unemployment rates.

